

Critical Analysis & Pathways Forward

State of research, interpretation quality, knowledge gaps, and next experiments

Thesis in one sentence

Seizure-frequency extraction is best understood as construction and projection of an auditable clinical frequency state - not as direct final-label prediction.

Prepared from the uploaded Gan 2026 Full Research Retrospective and the critical analysis drafted in conversation.

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A polished version of the critical research analysis, preserving the argument structure while adding visual organization for review and planning.

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Executive read

Bottom line

Your interpretation is mostly strong and unusually honest. The work has moved from 'which architecture wins?' to 'which component owns which clinical decision?' That is the right scientific reframing. The strongest contribution is not the current best metric; it is the emerging methodology: attribution layers, frozen deterministic comparators, evidence validity, coverage-versus-projection separation, and explicit validation/test caution.

The main weakness is that the current thesis has become plausible before it has become proven. In particular, 'state graph is the best path to generalization' is a strong working hypothesis, but the present evidence mostly shows that the state graph is a better diagnostic substrate - not yet that it yields a more generalizing final system.

Best current contribution

A modular experimental apparatus that can locate evidence, state representation, projection, repair, and rendering failures separately.

Largest current risk

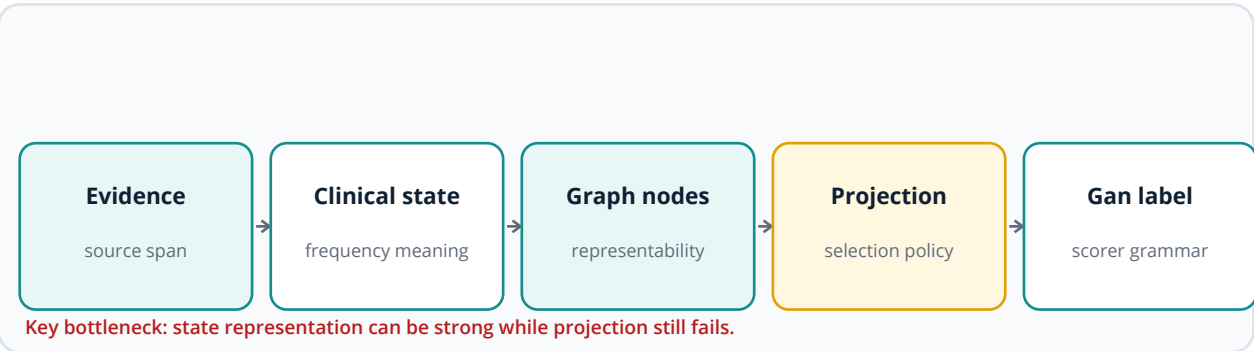
Validation-driven architecture drift: each known failure family prompts another module, creating complexity without proving transfer.

Most important bottleneck

Projection: choosing the clinically current, benchmark-relevant state from represented candidate states.

Highest-leverage next artifact

A hidden-family atlas with prevalence, failure ownership, and component-level metrics across validation, hard slices, and fresh audit rows.



Current research state	Interpretation	Confidence
Rules-only validation strength	Excellent as a transparent comparator, not as proof of generalization.	High
Structured-events threshold pass	A repair-heavy hybrid success rather than clean LLM-final-label ownership.	High
Hybrid adjudication	Candidate-recall-limited; useful result but not a promoted system.	High
State graph	Best diagnostic substrate and best research bet, but still needs fresh end-to-end proof.	Medium-high
LLM-heavy reasoner	Evidence-selection signal is real; clean final-label ownership remains unproven.	High

1. State of the research

The project is in a productive but dangerous middle state.

Why it is productive

You now have enough architecture diversity to see recurring invariants. Across rules, structured events, claim tables, adjudicators, state graphs, and LLM-heavy reasoners, the same pattern repeats: locating relevant text is easier than converting that text into the correct current clinical frequency state and then into Gan label grammar.

This strongly supports the central reframing: seizure-frequency extraction is a clinical state reconstruction problem rather than a simple classification or span-extraction problem.

Why it is dangerous

The repo now contains many development surfaces, score layers, repair modes, and architecture families. That creates a risk of epistemic diffusion: the project may appear to be learning a lot because it has many experiments, while some central questions remain under-isolated.

Recommended shift in success criteria

The next milestone should not be another aggregate validation score above 0.90. It should be: for each clinical subproblem, can the system show which component solves it, under what evidence constraints, with what regression risk, and on which distribution?

A concise diagnosis is: you do not yet have a final generalizing model, but you do have a strong research instrument for discovering why final models fail.

2. What the current interpretation gets right

2.1 Attribution discipline is a real strength

The retrospective correctly rejects metric-only claims. The structured-events result that reaches the validation threshold only after substantial deterministic repair is not a clean LLM-first success. It is a hybrid success, and the analysis should continue to say so.

This matters because many extraction systems quietly hide prediction-bearing logic inside 'normalization', 'repair', or 'format cleanup'. Your work has learned that repair layers are not neutral when they alter semantic answers. The distinction between shape-only repair and semantic repair should become one of the central methodological claims.

2.2 The rules-only comparator is being used appropriately

The retrospective correctly treats `rules_only_v1` as a frozen comparator rather than a production endpoint. Its validation performance is excellent, but its locked-test drop is too large to claim robust generalization.

Best use of rules now

The strongest role for rules is not 'rules beat LLMs'. It is: deterministic rules provide a controlled, inspectable baseline that exposes where LLMs add signal, where they regress, and where candidate generation fails.

2.3 Candidate recall is a true ceiling for constrained adjudication

The hybrid adjudicator result is scientifically useful because it shows a hard ceiling: when the correct answer is absent from the candidate set, an LLM restricted to choosing among candidates cannot recover it. That conclusion is sound for the constrained adjudicator design.

It should not be generalized too far. It proves that candidate-set-constrained adjudication is candidate-recall-limited. It does not prove that LLM adjudication is unhelpful, nor that an LLM should never

propose missing candidates.

2.4 The state-graph reframing is the central conceptual advance

The state graph is valuable because it separates questions that were previously confounded: Did we see the right evidence? Did we represent the right clinical state? Did projection choose the right represented state? Did benchmark label rendering fail? Did a repair layer change the answer?

The state graph should probably become the central research object. It should not yet be treated as the final claimed solution.

3. Where the interpretation may be overconfident

3.1 State graph is plausible, not yet demonstrated

The claim that the state graph is the best path toward generalization is a good research bet, but the evidence does not yet establish it as an empirical fact. The state graph has shown better decomposition, promising oracle coverage, useful boundary-node additions, duration-node replay, projection bottleneck localization, and limited invariance. It has not yet shown superiority on locked or fresh held-out final prediction.

The risk is that the state graph becomes rules-only v2 with better language: more structured and interpretable, but still tuned to known subfamilies. To avoid that, measure not only graph coverage but graph precision, sparsity, calibration, and projection stability.

3.2 LLM final-label ownership is underspecified

It is fair to conclude that the current prompted LLMs do not reliably own final Gan label rendering at scale. But the conclusion should be narrowed to the tested schema, prompting, and model setup. The model may be better at producing typed claims, temporal relations, uncertainty judgments, and candidate rationales than at emitting exact scorer-facing labels.

Better LLM question

Instead of asking whether the LLM can output the final label, ask which clinically meaningful decisions the LLM can make more robustly than deterministic rules under exact-evidence and regression constraints.

3.3 Prefix optimism needs a causal explanation

The retrospective correctly notes that validation prefixes were often too friendly. But prefix optimism is a symptom, not a cause. The project still needs to identify whether the cause is row ordering, template clustering, hidden-family imbalance, fewer temporal-boundary cases, scorer-format differences, or accidental similarity to development examples.

3.4 Aggregate locked-test knowledge can still shape decisions

The locked-test discipline is strong, but aggregate test knowledge can still exert research pressure. Knowing that candidate recall collapses on test can steer candidate-generation improvements even without row-level inspection. That is not necessarily wrong, but it should be acknowledged as test-informed research pressure.

The safest path is to introduce a new blinded audit set or fresh split before making strong final claims.

4. Core gaps in knowledge and reasoning

Gap	Why it matters	Diagnostic to add
<u>Hidden-subfamily ontology</u>	Current families are recognized qualitatively but not formally defined, labelled, and measured.	Create a multi-label atlas with prevalence, overlap, and per-component failure rates.
Causal error attribution	Wrong rows need the <u>first necessary failure point</u> , not just a broad failure category.	Tag evidence missing, evidence wrong, state not represented, projection wrong, rendering wrong, or gold ambiguous.
Graph precision cost	Oracle coverage can rise if the graph emits too many <u>conflicting nodes</u> .	Track node precision, contradiction rate, candidate count, and projection confusion.
Projection theory	Selection policy is spread across rules, repair, LLM prompts, and graph projection.	Write a <u>projection specification</u> with precedence rules and compare deterministic, LLM, and oracle projection.
Gold-standard ambiguity	Some notes may not imply a unique clinical answer; scorer grammar can penalize defensible states.	Run a clinician or <u>expert adjudication audit on hard and disagreement rows</u> .
Statistical uncertainty	Small aggregate deltas can mislead architectural conclusions.	Add paired tests, <u>bootstrap intervals by hidden family</u> , and correction/regression tradeoff plots.
Operational reliability	Cost, latency, schema failure, retries, and model-version sensitivity matter for clinical extraction systems.	Report LLM calls, cost per 1,000 notes, latency, retry rate, and reproducibility.

4.1 Hidden-family atlas is the biggest missing artifact

The retrospective repeatedly identifies hidden subfamilies: clusters, seizure-free duration, unknown/no-reference boundary, diary aggregation, recent-window arithmetic, competing semiologies, vague claims, template family, and scorer-format sensitivity. These should become a formal ontology with definitions, examples, prevalence, and inter-annotator reliability.

4.2 Causal attribution should replace broad error labels

For each wrong row, the analysis should assign the first necessary failure in the pipeline. A row may be wrong because evidence was missed, evidence was found but normalized incorrectly, the correct state was represented but not projected, the projection was right but Gan rendering failed, or the gold label was ambiguous. These are different scientific problems.

4.3 Graph representability is not enough

A graph that represents the gold state somewhere is useful, but only if projection can select it without oracle help. As node families grow, the graph may gain recall while losing precision or increasing contradiction. Coverage must be paired with precision and projection stability.

5. Potential pathways forward

Pathway	Central move	Promotion criterion
A. Hidden-family atlas	Annotate rows by clinically meaningful subfamilies and <u>report every architecture by family</u> .	Shows which phenomena cause the generalization gap and which component owns each fix.
B. State graph as IR	Treat the <u>graph as a true intermediate representation</u> with its own evidence, precision, coverage, sparsity, and projection metrics.	Graph improvements transfer without hidden candidate explosion or projection regressions.
C. Projection benchmark	Separate node selection from node construction and compare <u>deterministic policy</u> , LLM projection, pairwise ranking, hybrid abstention, and oracle projection.	Projection gains appear on broad and fresh surfaces with bounded regressions.
D. Candidate generation revisit	Let the LLM propose missing exact-evidence candidates, then verify and normalize them before adjudication.	Candidate recall improves on hard families without candidate explosion or precision collapse.
E. Complementarity ensemble	Use LLMs selectively where they fix deterministic misses with high precision.	LLM overrides have positive selective-action precision and low regression rate.
F. Abstention	Add <u>underdetermined, conflicting, stale-only, or human-review outputs</u> and report coverage-accuracy curves.	System identifies uncertainty rather than forcing unsafe labels.
G. Gold audit	<u>Adjudicate hard and disagreement rows</u> for ambiguity, benchmark convention issues, and possible gold errors.	Clarifies achievable ceiling and separates model failure from label ambiguity.
H. Typed LLM reasoning	Ask LLMs for typed operations and source-near claims, then use deterministic compilers for arithmetic and Gan grammar.	Preserves LLM clinical signal while avoiding hidden semantic repair.

5.1 Build the hidden-family atlas before another architecture

This should be the highest-priority next step. Each row should receive binary or multi-label tags such as explicit rate, seizure-free duration, seizure-free boundary with residual events, diary/log aggregation, cluster count, recent-window arithmetic, vague frequency, competing semiologies, historical-only frequency, template shorthand, scorer-format edge case, and gold ambiguity.

The output should be a family-level matrix showing prevalence, rules recall, LLM evidence recall, graph coverage, projection accuracy, and main failure owner. That table would be more valuable than another aggregate model score.

5.2 Turn the state graph into a true intermediate representation

For each note, the graph should expose source evidence, assertion status, temporality, semiology, frequency quantity, denominator/window, cluster/diary arithmetic, seizure-free duration, uncertainty type, contradiction links, projection priority, and final Gan rendering.

Evaluate it through separate metrics: node evidence precision, gold-state representability, graph parsimony, projection accuracy given graph, rendering accuracy given projected state, and final end-to-end score.

5.3 Separate projection into its own research problem

Projection is now the bottleneck and should be benchmarked directly. Compare deterministic precedence policies, pairwise node rankers, LLM projection over exact-evidence graph nodes, hybrid projection with abstention, and oracle projection upper bounds.

High-value experiment

Compare deterministic nodes + deterministic projection, deterministic nodes + LLM projection, deterministic + LLM boundary nodes + deterministic projection, deterministic + LLM boundary nodes + LLM projection, and oracle projection. This directly tests whether LLMs help more in node creation or node selection.

5.4 Revisit candidate generation rather than only adjudication

The hybrid adjudicator may have discouraged LLM adjudication too broadly. The real issue may be that the LLM was not allowed to propose missing candidates. Test a three-stage candidate system: deterministic candidates, LLM missing-candidate proposer with exact-evidence requirement, verifier-normalizer, and adjudication over the union.

5.5 Use complementarity instead of replacement

Claim-table v5 is weak as a replacement but strong as evidence of complementarity. A selective ensemble should identify high-confidence rules rows, identify productive disagreements, route only a subset to LLM override, and abstain where disagreement lacks strong support.

The key metric becomes selective-action precision: when the LLM changes the deterministic answer, how often is that change correct?

5.6 Treat abstention as a first-class output

A clinical extraction system should know when a note is underdetermined. Evaluate final label, unknown frequency, insufficient evidence, conflicting evidence, stale-only evidence, and human-review outputs. Report accuracy at different coverage levels rather than only forced-label F1.

5.7 Audit the gold labels and scorer conventions

A hard-row adjudication study would distinguish model failure from benchmark ambiguity. For rows where all systems fail, rules and LLMs disagree, exact selected evidence still yields the wrong label, or predictions seem clinically plausible but scorer-rejected, classify whether the gold is uniquely implied, clinically defensible alternatives exist, benchmark convention dominates, note is underdetermined, or gold is likely wrong.

5.8 Reframe LLM-heavy work as typed reasoning

The current LLM-heavy failure is not that the model cannot find useful evidence. It often can. The failure is that raw final labels remain too unreliable. The next LLM-heavy target should be typed operations: event count, time window, denominator, cluster size, seizure-free duration, temporal anchor, semiology grouping, uncertainty type, and selected evidence ID.

The clean architectural claim would be: the LLM owns source-near clinical interpretation; deterministic code owns arithmetic and benchmark compilation.

6. Pathways that may have been incorrectly ruled out

Pathway	Why it remains plausible	How to test it safely
LLM as missing-candidate generator	Constrained adjudication could not recover absent candidates, but that does not rule out LLM candidate recall expansion.	Require exact evidence, verifier-gate candidates, measure recall/precision/candidate explosion.
Pairwise ranking	<u>Direct final selection may be harder than pairwise comparison of two candidate states.</u>	Run tournaments over <u>graph nodes</u> and compare against deterministic and oracle projection.
Probabilistic projection	Many notes support degrees of confidence rather than a single obvious state.	<u>Preserve confidence over states and report calibration</u> plus abstention curves.
Retrieval/exemplar routing	Hidden template families may benefit from nearest-neighbor routing or family-specific examples.	Use only fixed retrieval policies and evaluate on fresh/blinded panels to avoid memorization.
Human-in-the-loop endpoint	Clinical usefulness may come from exact evidence and uncertainty flagging over forced labels.	Measure coverage, triage precision, review burden, and corrected final accuracy.
Benchmark compiler module	Gan grammar is distinct from clinical interpretation and should not contaminate extraction.	Compile typed clinical states into Purist, Pragmatic-equivalent, and scorer-facing labels.

The key theme is that negative results should be interpreted at the level of the tested role, not the whole architectural family. For example, candidate-set-constrained LLM adjudication failing does not rule out LLM missing-candidate generation, LLM projection, or typed LLM reasoning.

7. Recommended next research sequence

- 1 Stop broad architecture proliferation temporarily. You already have enough architectures to support the central research claims; the bottleneck is causal understanding.
- 2 Build the hidden-family atlas across validation, hard slices, disagreement rows, and a fresh blinded sample if available.
- 3 Perform causal row-level failure decomposition with the first necessary failure point for each wrong row.
- 4 Make projection a formal benchmark using graph nodes, deterministic projection, LLM projection, pairwise ranking, and oracle projection.
- 5 Test LLM missing-candidate generation under exact-evidence and verifier-gated constraints.
- 6 Add selective prediction and report forced-label accuracy alongside coverage-accuracy curves.
- 7 Run a fresh audit only after component hypotheses, projection policy, and candidate-generation changes are fixed.

Near-term decision rule

Do not promote a component because it fixes a known synthetic or prefix panel. Promote only when it improves a predeclared hidden-family target while preserving broad regression panels and attribution clarity.

8. Revised thesis and risk register

Revised thesis statement

Proposed thesis

Seizure-frequency extraction is best understood as construction and projection of an auditable clinical frequency state, not as direct label prediction. LLMs and deterministic rules both contribute, but their contributions must be separated into evidence discovery, state representation, projection, arithmetic, and benchmark rendering. Apparent aggregate gains are unreliable unless each prediction-bearing layer is attributed, ablated, and tested under hidden-family distribution shift.

Risk register

Risk	Failure pattern	Mitigation
Validation-driven architecture drift	Known failures keep producing new modules that improve narrow panels but not transfer.	Predeclare hidden-family targets and regression panels.
Oracle-coverage optimism	The graph contains the right answer somewhere but projection cannot select it.	Always pair coverage with projection and graph precision metrics.
Semantic repair laundering	Repair or benchmark alignment does prediction work while being described as cleanup.	Report raw, format-only, selected-evidence arithmetic, benchmark-aligned, fallback, and full-stack layers separately.
Understudied ambiguity	Clinically defensible alternatives are treated as plain model errors.	Run gold/scorer ambiguity audits on hard and disagreement rows.
LLM role overgeneralization	A failure in one role is used to reject a different possible LLM role.	Evaluate LLM roles separately: extractor, proposer, ranker, projector, compiler.

9. Final assessment

The research is in a strong position intellectually, but not yet in a strong position as a final system. The interpretation is unusually careful and mostly correct: final-label F1 is too lossy, repair attribution matters, candidate recall limits constrained adjudication, and state representation is the right abstraction.

The next leap should be methodological rather than architectural: formal hidden-family stratification, causal error attribution, graph precision and coverage metrics, projection-only benchmarking, LLM missing-candidate generation, selective abstention, and gold ambiguity audit.

The winning move is probably no longer 'try another prompt'. It is to turn the current system into an experimental apparatus that can prove which clinical decision each component owns, where it fails, and whether that ownership transfers.

Source note

This PDF is based on the uploaded `gan2026_full_research_retrospective_2026-06-02.md` and the critical analysis generated from it in this conversation. The uploaded retrospective states that the work is a research retrospective, not a benchmark-comparison claim; it frames validation results as development evidence and locked-test results as aggregate/frozen-audit context only.